

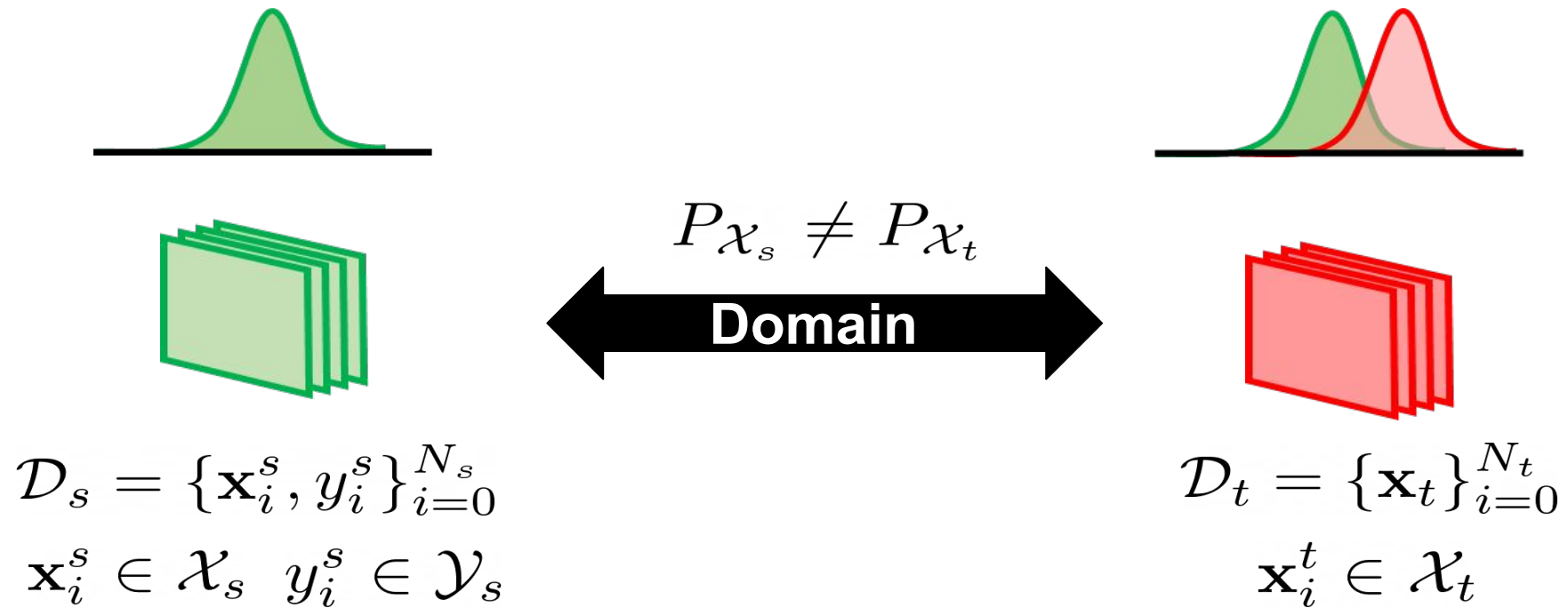
TTTFlow: Unsupervised Test-Time Training with Normalizing Flow

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*equal contribution

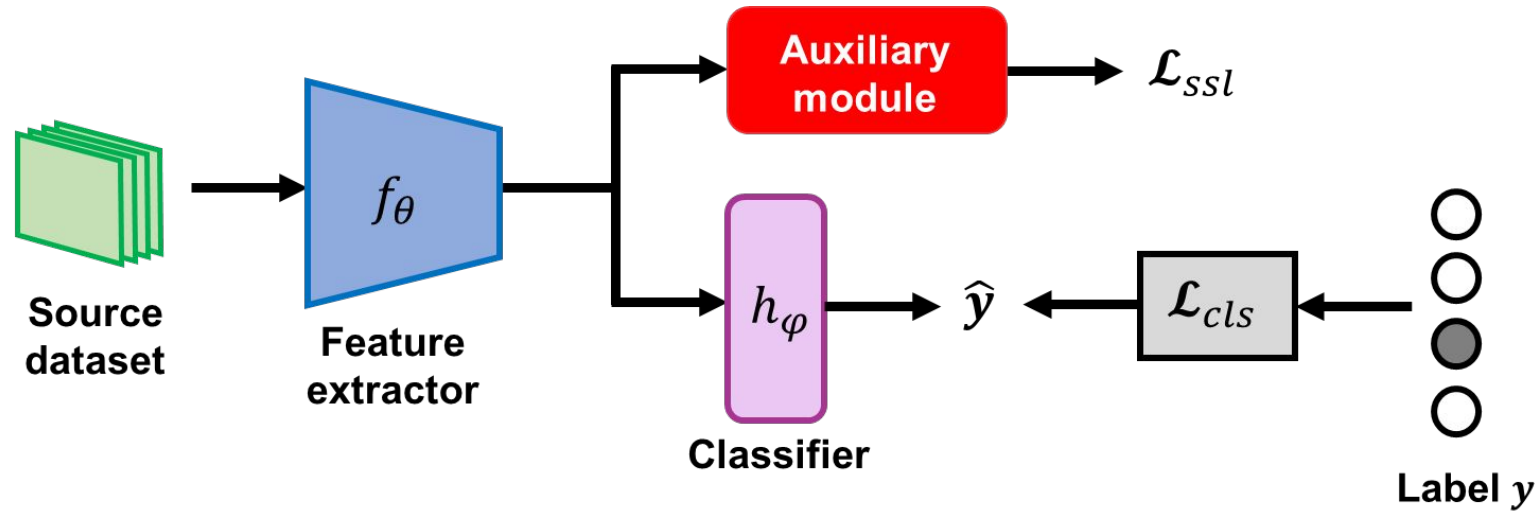
Test-Time Adaptation

In Test-Time Adaptation, a model is adapted to a target dataset without accessing the training (source) data:



Test-Time Training

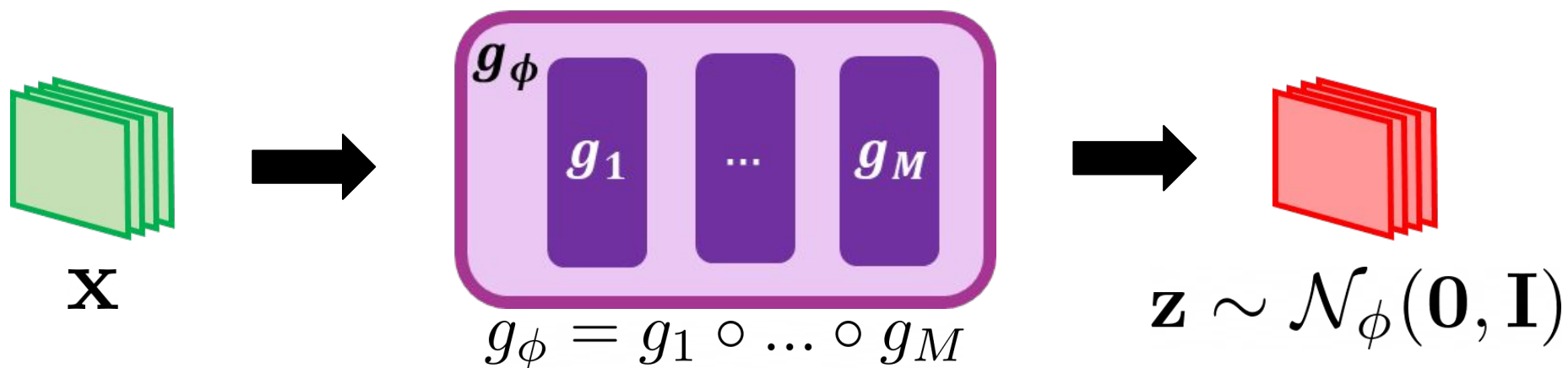
An auxiliary task is used at training time and at test-time:



Examples: TTT (rotation prediction), TTT++ (contrastive learning)

Normalizing Flows

Normalizing Flows transforms data from a complex and unknown distribution, to a well-defined and tractable distribution:

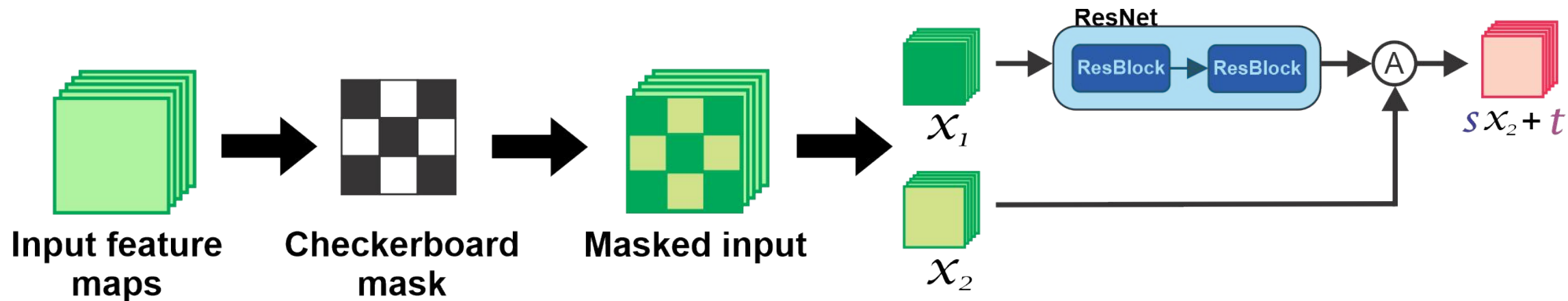


We use the change of variable rule:

$$\log P(\mathbf{x}) = \log P_\phi(\mathbf{z}) + \log \left| \det \left(\frac{d\mathbf{z}}{d\mathbf{x}} \right) \right|$$

Normalizing Flows

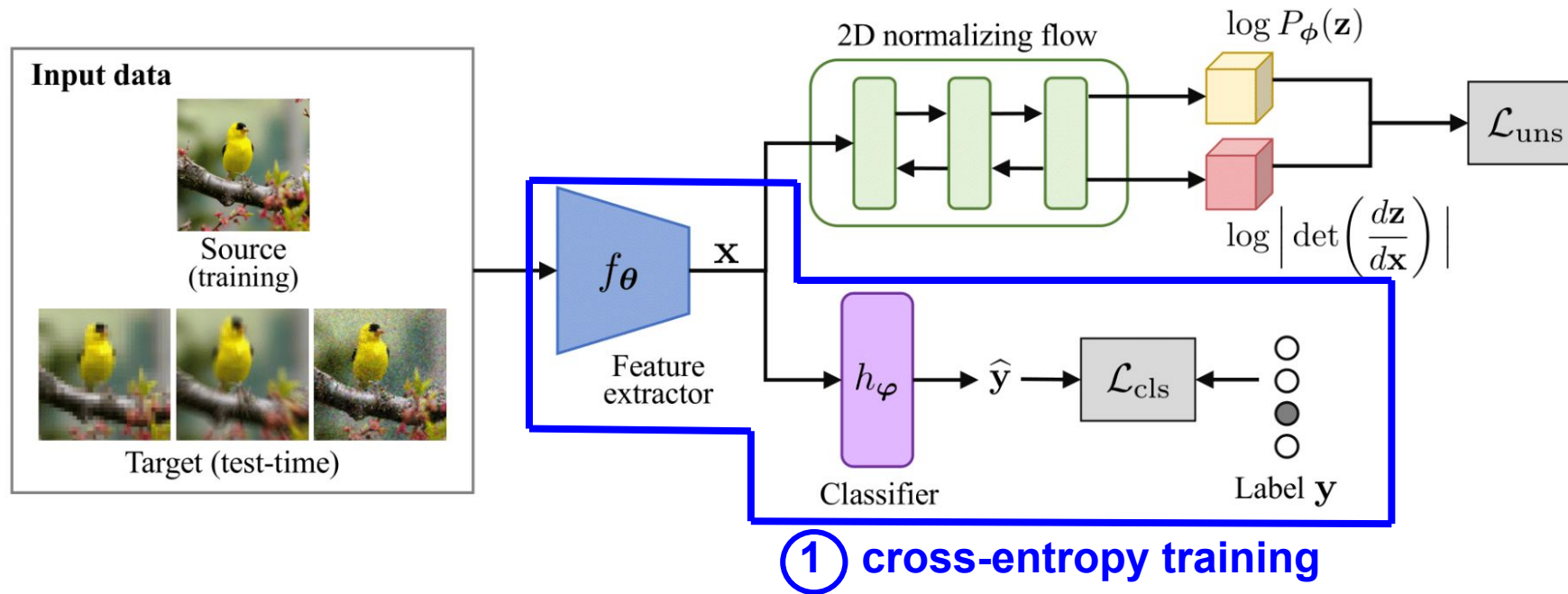
We use affine coupling layers from RealNVP:



TTTFlow: a Normalizing flow learns the mapping from the pretrained model's feature maps.

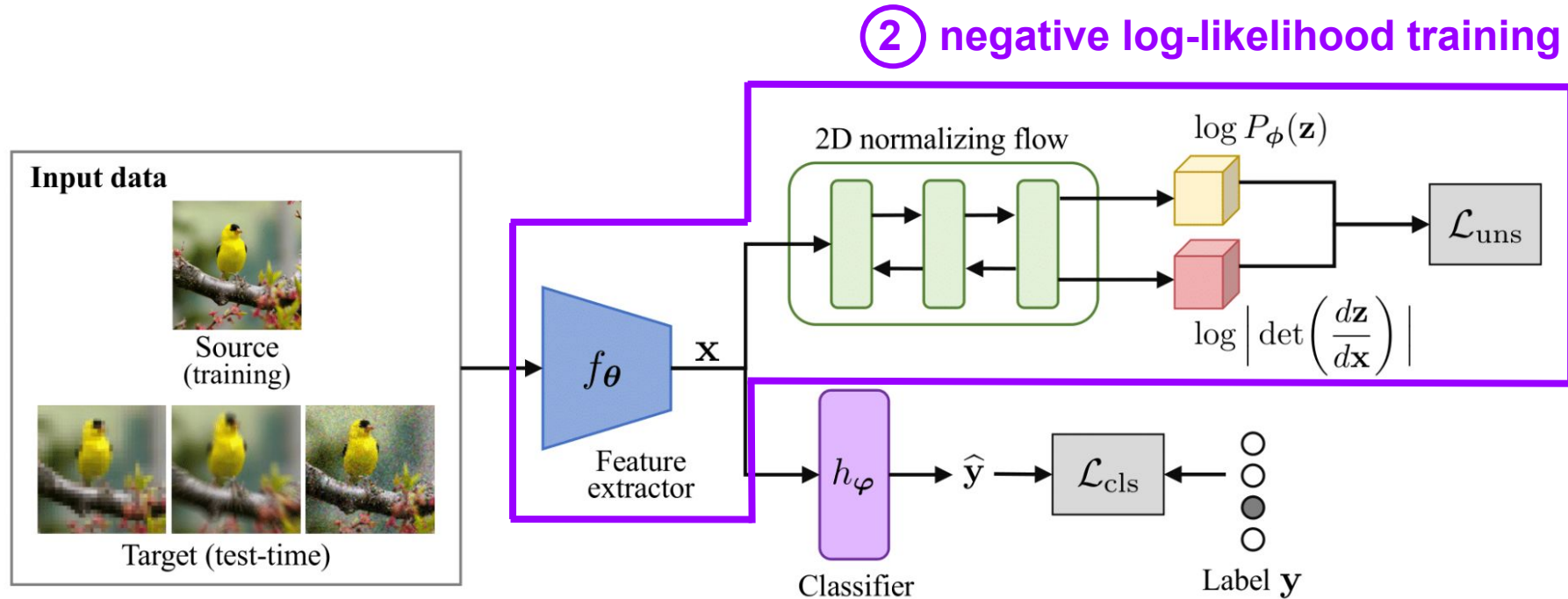
TTTFlow

We introduce Normalizing Flow as an unsupervised task:



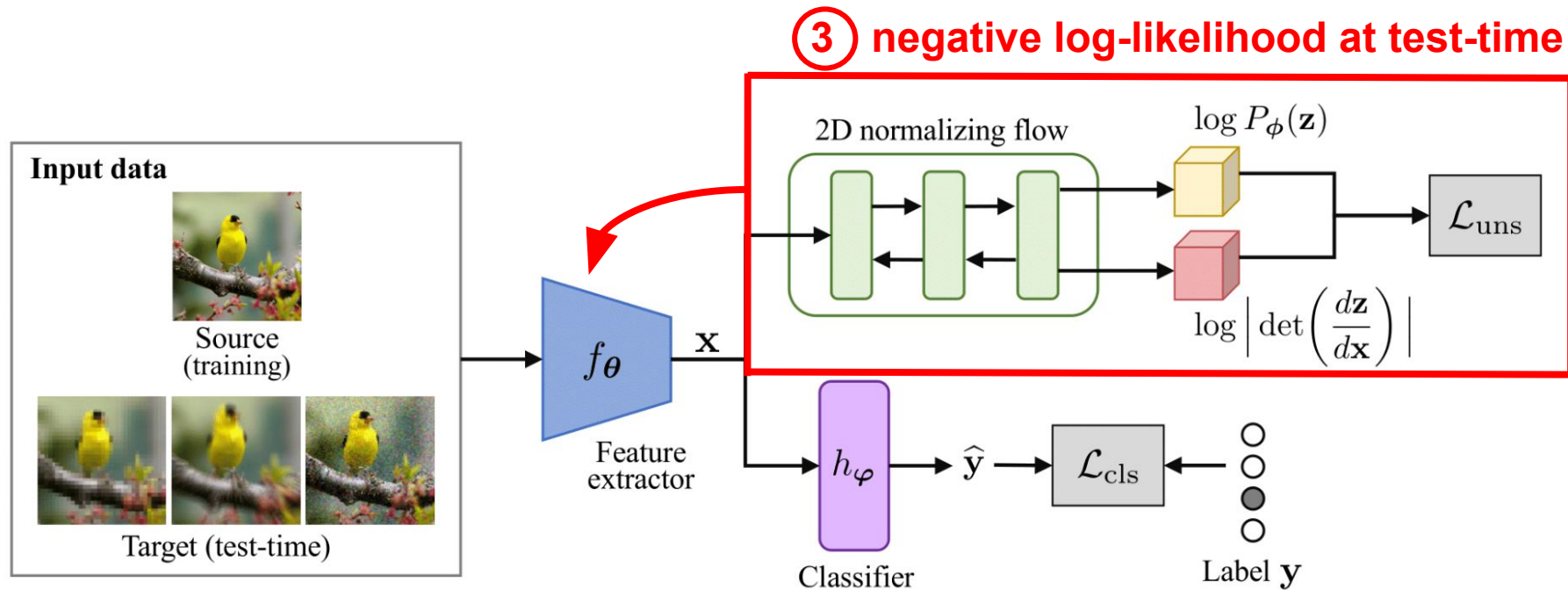
TTTFlow

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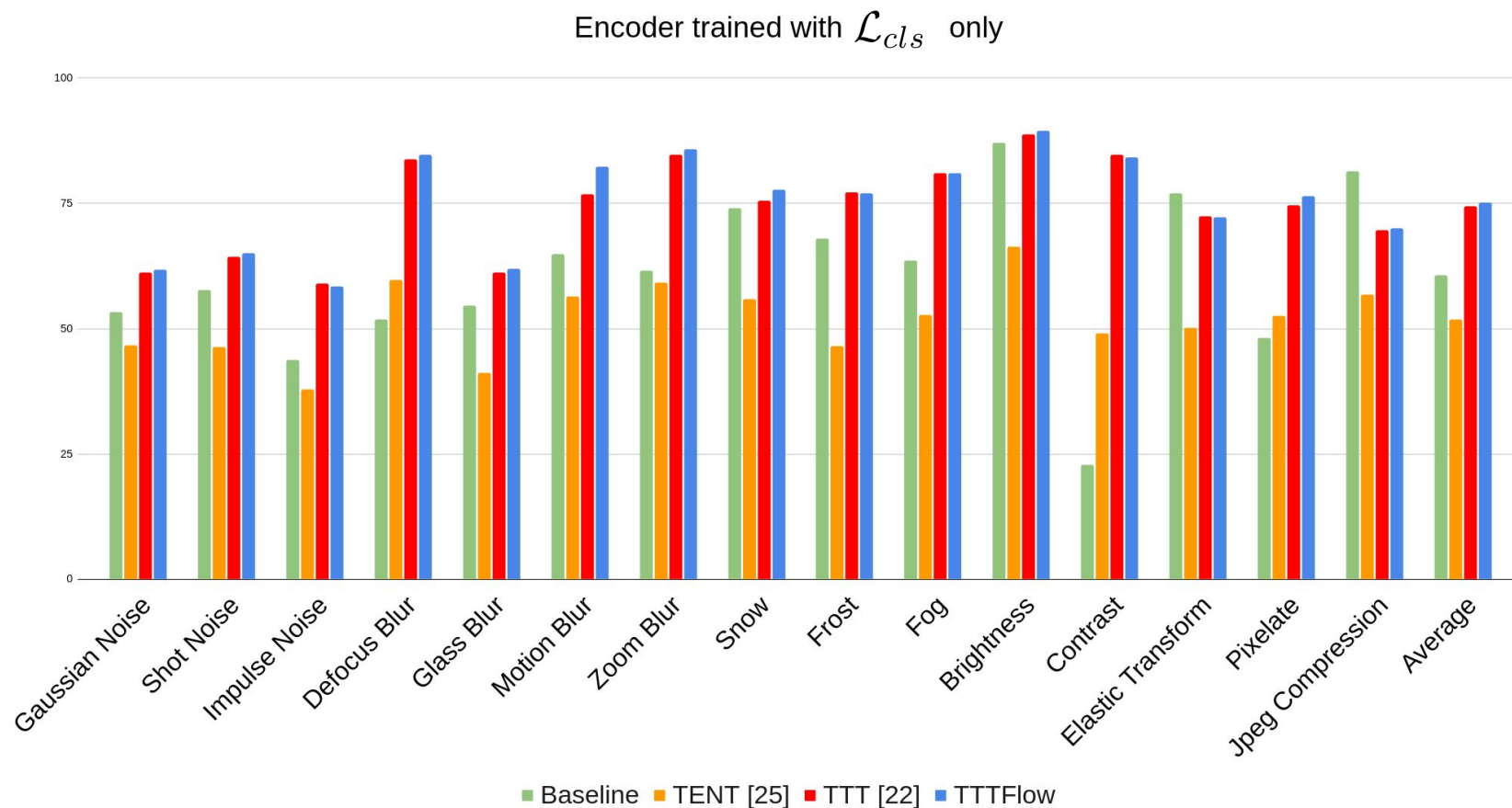
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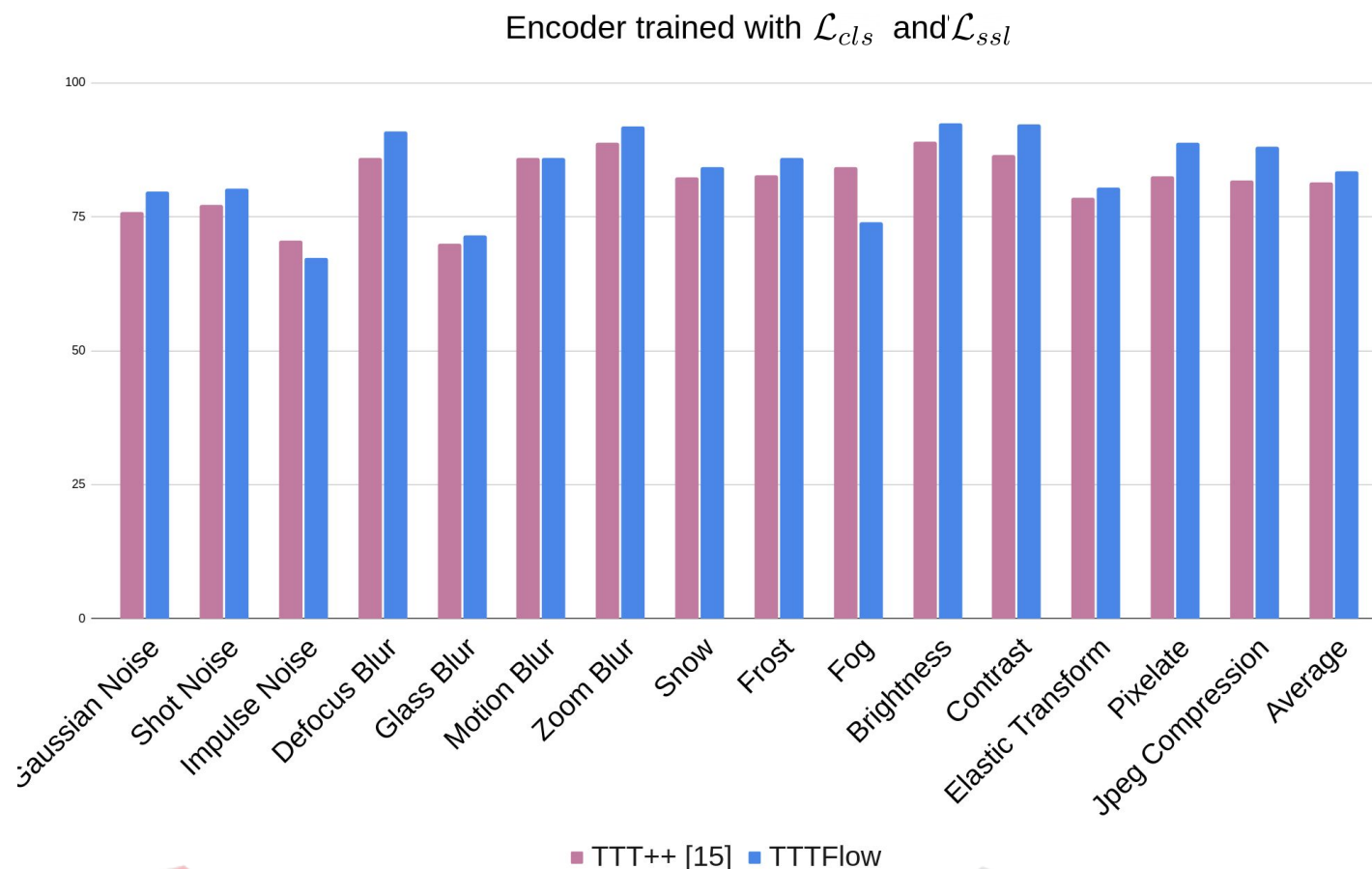
Object recognition on corrupted images

Experiments on the CIFAR-10-C dataset trained only with $\mathcal{L}_{cls}$:



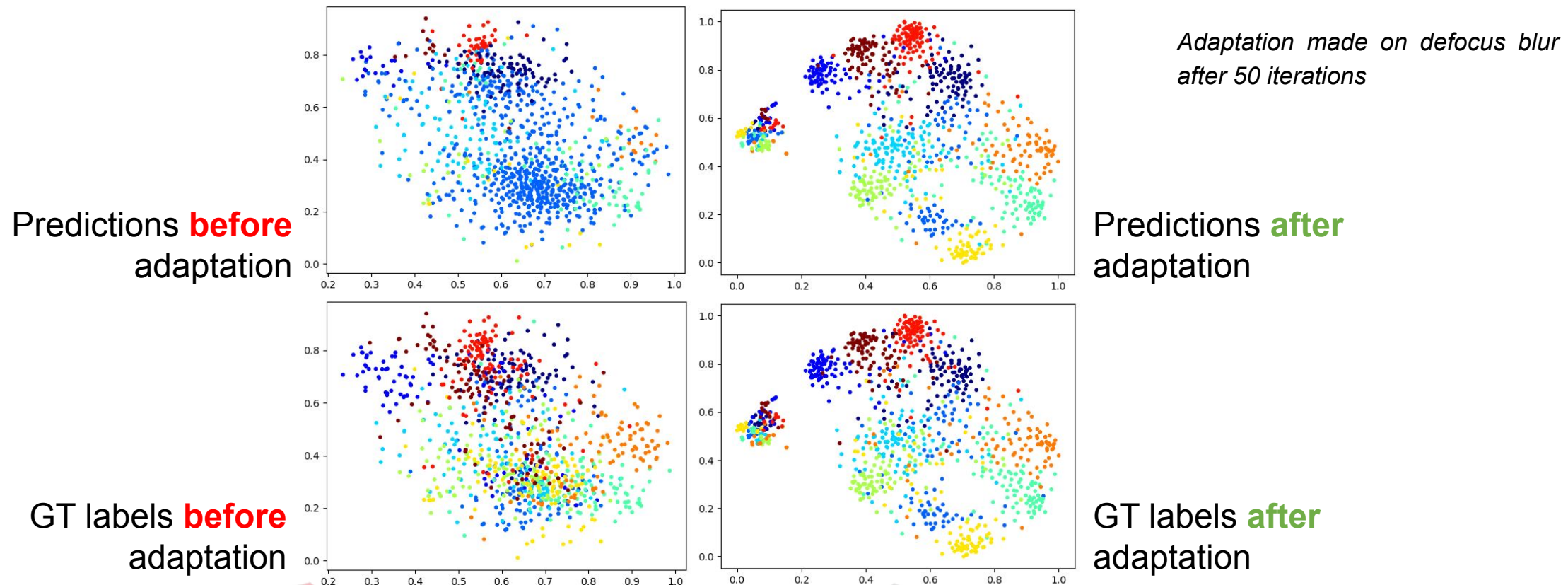
Object recognition on corrupted images

Experiments on the CIFAR-10-C dataset trained with $\mathcal{L}_{cls} + \mathcal{L}_{ssl}$:



Object recognition on corrupted images

t-SNE plots of features at the end of extractor, before and after adaptation:



Object recognition on natural domain shift

Experiments on the CIFAR-10.1 dataset:

Method	Accuracy (%)
Baseline	84.70
TENT [25]	58.98 $\pm$ 0.12
TTT [22]	84.49 $\pm$ 0.15
TTTFlow (Lcls)	85.11 $\pm$0.30
TTT++ [15]	88.24 $\pm$0.17
TTTFlow (Lcls + Lssl)	86.49 $\pm$ 0.02

Contact us if you have any questions:

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And look for the code and the whole article here:

